



RESEARCH ARTICLE

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Key Points:

- Geomagnetically induced currents (GICs) are estimated for mainland Portugal using the first 3D crust-mantle conductivity model for the whole territory
- The geotectonic environment strongly influences the location of power network substations most susceptible to GICs
- First GIC measurements at Paraimo substation are used to test the conductivity model

Supporting Information:

Supporting Information may be found in the online version of this article.

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Toward More Accurate GIC Estimations in the Portuguese Power Network

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Abstract New geomagnetically induced current (GIC) computations for mainland Portugal include the entire power network, with network parameters and topology provided by the transmission grid operator for all the high voltage lines (150, 220, and 400 kV). The first 3D conductivity model for the west region of the Iberian Peninsula, based on 31 broadband magnetotelluric soundings, is used in calculations, revealing the effect of different crustal domains in GIC distribution. Geomagnetic field variations are taken from Coimbra or San Fernando magnetic observatories, according to the Nearest Neighbor method, and used together with surface impedance values predicted from the new conductivity model to calculate the induced electric field on a regular grid. The global distribution of GICs over the power network is characterized based on results derived for the eight most significant storms registered in the Iberia during solar cycle 24. Substations susceptible to the highest GICs are found near the transition between the granitic geotectonic unit of Central Iberian Zone and the Lusitanian Basin. A prototype of a Hall effect sensor has been installed at a substation and is active since the end of August 2021. In order to validate our GIC model, recent measurements are compared with simulations. GIC computation is prone to uncertainties from various sources, possibly contributing with different weights to the final error in computed values. Here, we evaluate the contribution of substation earthing resistance and nonuniqueness of the conductivity model to the final GIC uncertainties.

Plain Language Summary Magnetic storms on the Sun can trigger geomagnetic disturbances (GMD) on Earth, affecting technological infrastructures and gadgets we depend upon. As an example, they can interfere with the normal operation of power networks transporting electricity by inducing spurious currents along the conducting lines, which are called geomagnetically induced currents (GICs). We computed GICs at all substations of the transmission power network in Portugal mainland, gathering information on the electric circuit parameters, the most intense GMD events registered at the magnetic observatories of Coimbra (Portugal) and San Fernando (Spain), and measuring the Earth response to electromagnetic waves of different frequencies to obtain a conductivity model. Our results show the influence of local geotectonic structures on the geographic distribution of GICs during a given GMD. GICs tend to be stronger close to sharp discontinuities of the ground resistivity, which occur at the transition between different geotectonic units. Among different possible sources of errors in GICs, we compute error estimates due to uncertainties in the ground conductivity model and in the grounding resistances at certain substations. Finally, we give account of a first GIC sensor installed in the power network, and use its data to test different conductivity models.

1. Introduction

Geomagnetically Induced Currents (GICs) are a result of intense and sudden changes in the Earth's magnetic field due to variations in the ionospheric and magnetospheric electric currents when energetic particles and/or electromagnetic radiation from the Sun reach the Earth's environment. These so-called geomagnetic disturbances (GMD) or geomagnetic storms induce electric fields in the Earth's crust and mantle. The induced electric field drives electric currents in closed circuits along grounded conductive structures. Extreme GICs represent a threat to man-made infrastructures such as power grids, pipelines, telecommunication cables, and railway systems (e.g., Baker et al., 2008; Boteler, 2021; Boteler & Pirjola, 2017; Molinski, 2002; Pirjola et al., 1999).

In power lines, depending on the severity of geomagnetic storms, geographical location, the topology of the grid, and electrical properties of the underlying ground, significant GICs can be produced (Viljanen et al., 2014). As a result, multiple adverse effects may occur, ranging from transformer heating, harmonic generation, and

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voltage fluctuations to power blackouts and permanent damage to transformers (see, e.g., Pirjola et al., 1999; Molinski, 2002; Thornberg, 2012). The most known dramatic GIC impact concerns the March 1989 event, when a geomagnetic storm caused the collapse of the Hydro-Québec power grid system in Canada, leading to a 9 hr blackout (Guillon et al., 2016). Since then, GIC hazard mitigation has been considered important for planning and designing more resilient transmission systems.

An increasingly large number of studies have focused on this subject during the past decades. Mid-latitude power networks are also considered at risk, as can be concluded from the report of important failures that took place in South Africa (Gaunt & Coetzee, 2007) and New Zealand (Marshall et al., 2012). In the framework of project MAG-GIC (<https://www.uc.pt/en/org/maggic>), the first simulations for the Portugal mainland power network (Alves Ribeiro et al., 2021) used a simplified 3D conductivity model for the southern region. That is, lateral heterogeneities in electrical conductivity were included through ocean-land conductivity contrasts and concerned only the upper 4–5 km of crust. In order to obtain more realistic conductivity models to use in simulations, a field measurement campaign has been carried out to complete a 50×50 km grid of magnetotelluric (MT) soundings over the whole territory. The MT method is a passive electromagnetic geophysical method that measures the variations of the natural electric and magnetic fields at the surface of the Earth (Cagniard, 1953; Tikhonov, 1950). From the gathered data set, and after selection tests were applied, an empirical 3D conductivity model for the whole territory was inverted (Baltazar-Soares et al., 2020, 2021). In this study, we use Baltazar-Soares et al. (2021) (B-S) conductivity model to simulate GICs in the global Portuguese power network. Furthermore, the national transmission system operator, REN (Redes Energéticas Nacionais), has provided most of the required circuit parameters, such as the location of substations and their grounding resistances, the characterization of each transformer and information about the transmission lines regarding resistance and length values. For substations that belong to REN (REN substations), we considered the provided grounding resistances as exact values. However, in the case of non-REN substations, a standard grounding resistance of 0.2Ω was assumed by default, and associated uncertainties were estimated from differences to exact values of neighboring REN substations.

The new GIC model was run to compute the induced currents due to the eight strongest geomagnetic storms of solar cycle 24, ordered according to the local K-index at the magnetic observatory of Coimbra (COI). Geomagnetic field observations are crucial for the computation of geoelectric fields that induce GICs and are not exactly the same for the entire territory, whose northmost and southmost points are located at geomagnetic latitudes 45.4° and 40.3° , respectively. Different approaches can be used to interpolate the geomagnetic field, including the Spherical Elementary Current System inversion (SECS) (Amm & Viljanen, 1999), the Spherical Cap Harmonic Analysis (Haines, 1985) and purely empirical methods such as the Nearest Neighbor, the Triangle Barycentric Interpolation and the Gaussian Process Regression (Rigler et al., 2019). For most of the territory, the Coimbra magnetic observatory is the nearest source of continuous measurements. The only exception is Portugal's southernmost region, where substations are closer to the San Fernando observatory (IAGA code SFS) than to COI. The largest latitudinal distance of a given substation to the closest between the two magnetic observatories is always less than 2.0° , so we considered that the Nearest Neighbor method could be used. This approximation is in line with previous studies where this method presented as good performances as using more complex methods, at geomagnetic dipole mid-latitudes (Rigler et al., 2019).

Since the end of August 2021, direct measurements of low-frequency currents in the transmission power grid have been carried out to validate our GIC model. In collaboration with colleagues working on developing instrumentation (R. R. Santos, 2021), a prototype for an acquisition device to measure GICs has been assembled, calibrated, and tested in the laboratory, and then installed at a substation about 35 km from Coimbra, called Paraimo (SPI). The instrumental setup relies on a Hall effect sensor LEM HOP 1000-SB, clamped to the neutral grounded cable of transformer TRF6. It was built over a single-board computer Raspberry Pi 4, with a high-resolution digitizer (24-bits).

Sensitivity tests regarding GIC simulations have shown that results may significantly depend on circuit parameters not known with great precision as the grounding resistance at certain substations and, above all, on uncertainties in the conductivity model (e.g., Beggan, 2015; Marshall et al., 2019). The assessment of errors in GIC computations due to the lack of precise information on those two items is also carried out.

Sections 2 and 3 introduce the geomagnetic source signal and the conductivity model for this study. Then, GIC estimations for the eight strongest events of the solar cycle 24 are shown in Section 4. Section 5 compares simulations with measurements at the Paraimo substation, and Section 6 discusses the effect of uncertainties in the

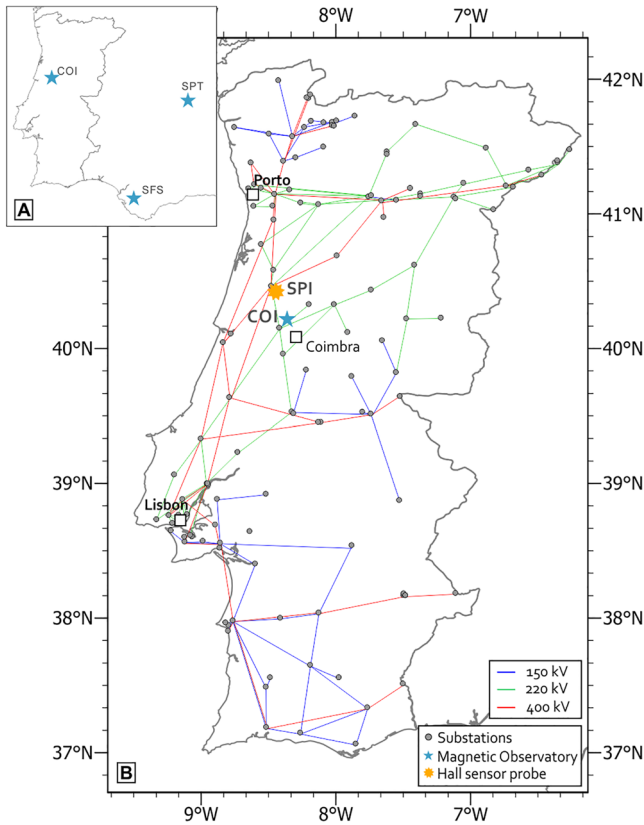


Figure 1. (a) Location of magnetic observatories closest to the power network (COI: Coimbra; SPT: San Pablo-Toledo; SFS: San Fernando); (b) Location of the substations for the high voltage power lines (150, 220, and 400 kV); the SPI substation (orange star) is where the geomagnetically induced current sensor is installed.

conductivity model and in the substations grounding resistance. Final discussion and conclusions are drawn in Section 7.

2. Geomagnetic Source Signal

The geomagnetic observatories closest to the Portuguese power network are those from Coimbra (COI), San Pablo Toledo (SPT) and San Fernando (SFS), as shown in Figure 1, with SPT at a distance about 380 km to the east and SFS about 460 km to the south-east of COI. Differences between the geomagnetic field time series are larger when comparing COI and SFS than COI and SPT. These results agree with the Ngwira et al. (2009) study, where it is shown that GIC source gradients are higher along the south-north than along the west-east directions. The geomagnetic time series for the south-north and west-east components ($X(t)$ and $Y(t)$, respectively) are used to compute the geomagnetically induced electric field in the grid of points where the surface impedance tensor is provided. The Nearest Neighbor method was applied in each one of the grid points, and the electric induced field was computed using the geomagnetic storm signal as observed at the nearest geomagnetic observatory among COI and SFS.

In this study, we focused on the most intense geomagnetic storms of solar cycle 24 (Table 1 and Figure 2), whose origin were coronal mass ejections (e.g., Watari, 2017). To characterize the magnitude of GICs computed for a given storm, we chose the standard deviation statistic, $SD(GIC)$, as this represents the level of GIC activity during the whole storm duration (see also Alves Ribeiro et al., 2021):

$$SD(GIC) = \sqrt{\frac{\sum_i (GIC(t_i) - \overline{GIC})^2}{N - 1}} \quad (1)$$

with N the number of 1-min values and $\overline{GIC}$ the average value in the storm time interval. This choice, instead of GIC peak values, is due to the fact

that the latter can be erroneously identified from outliers and also because GIC peak values are not necessarily simultaneous over the whole power grid (Girgis et al., 2020). A further reason is that, during a geomagnetic storm, the geomagnetic field as well as the corresponding electric induced field, change their polarization (alignment direction). Certain directions of the electric induced field are prone to larger GICs than others (e.g., Boteler

Table 1

List of the 8 Most Significant Geomagnetic Storms From Cycle 24 Used in This Study: Maximum Values of the K_p and Local K_{COI} Indices, As Well As Peak and Standard Deviation Values for Time Derivatives of the Horizontal Geomagnetic Field Components (as Measured at COI)

Storm	min(Dst) (nT)	K_p	K_{COI}	Max $ dH/dt $ (nT/min)	Max $ dX/dt $ (nT/min)	Max $ dY/dt $ (nT/min)	SD(dH/dt) (nT/min)	SD(dX/dt) (nT/min)	SD(dY/dt) (nT/min)
17/03/2015	-234	8-	6	34.4	25	25	1.59	1.34	1.48
22/06/2015	-198	8+	8	58.24	56	16	1.54	1.5	1.01
20/12/2015	-166	7-	7	19.31	18	13	1.16	1.01	1.05
09/03/2012	-145	8	6	21.59	21	11	1.47	1.47	1.21
08/09/2017	-122	8+	6	35.9	35	15	1.69	1.48	1.26
05/08/2011	-115	8-	6	27.89	27	11	1.36	1.25	0.92
05/04/2010	-81	8-	6	20.8	17	12	1.89	1.14	1.74
02/10/2013	-72	8-	6	36.23	23	28	1.68	1.33	1.32

Note. For a quantity f , $SD(f)$, is computed using Equation 1.

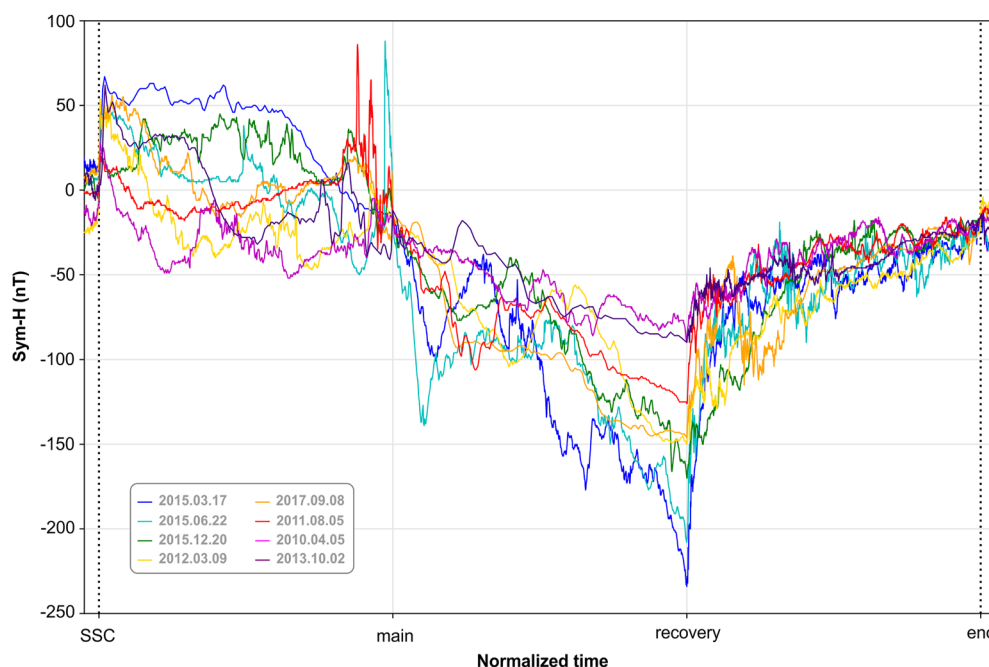


Figure 2. Sym-H index during different stages of the geomagnetic storms displayed in Table 1. Sudden storm commencements are located according to the 2007 XXIV IUGG General Assembly (Curto et al., 2007), while the remaining phases are located following the criteria of Walach and Grocott (2019) and Hutchinson et al. (2011). The time scale is different for different storms, so that corresponding storm phases have the same (normalized) duration.

& Pirjola, 2017), so the SD(GIC) value will characterize better the overall GIC-effectiveness of the geomagnetic storm field polarization.

The definition of the geomagnetic storms' duration is required to compute SD(GIC), and this topic lacks consensus (Haines et al., 2019). In fact, several groups use different criteria based on geomagnetic indices such as the Dst (Yokoyama & Kamide, 1997), SYM-H (Hutchinson et al., 2011; Walach & Grocott, 2019) or the *aa* index (Haines et al., 2019). Sudden storm commencements (SSCs) are often precursors of geomagnetic storms (Curto et al., 2007; Mayaud, 1973). Our study used the SSC tables provided by the Ebre Observatory (<https://www.obsebre.es/en/rapid>) for the beginning of each storm, where the criteria to identify SSCs are those defined at Perugia's 2007 XXIV IUGG General Assembly (Curto et al., 2007). According to them, a SSC is detected when there is a 3 nT/min (or larger) variation of the magnetic field followed by an increase of the geomagnetic activity ($Dst \leq -50$ nT) within the following 48 hr. On the other hand, to locate the end of each storm, we used the same criteria as Hutchinson et al. (2011), that is, the instant when the SYM-H index reaches the -15 nT quiet level while recovering from more negative values. Between the two end points of a geomagnetic storm, and according to Walach and Grocott (2019), the storm's main phase takes place in the period between the instant when the -15 nT level of SYM-H index is crossed while decreasing and the instant when the SYM-H minimum is attained; as to the recovery phase, it begins when the minimum SYM-H value is reached.

In Figure 2, the 8 most intense geomagnetic storms of solar cycle 24 are plotted. A normalized time scale is used for easier comparison among different storms, whereby the onset and offset of each storm phase (initial, main, recovery) are made to coincide. However, in computing SD(GIC), the same time scale is used for all storms.

3. 3D Conductivity Model for Portugal Mainland

The magnitude and orientation of the induced electric field $\mathbf{E}$ strongly depend on the subsurface conductivity structure (Bedrosian & Love, 2015) and, for this reason, a precise conductivity model is required. The 3D conductivity model for more accurate GIC estimations was calculated from 31 MT soundings acquired with a site spacing of approximately 50×50 km (Baltazar-Soares et al., 2021) (Figure 3), and will be called B-S conductivity model in the following. Eight MT soundings located in the south region of Portugal were

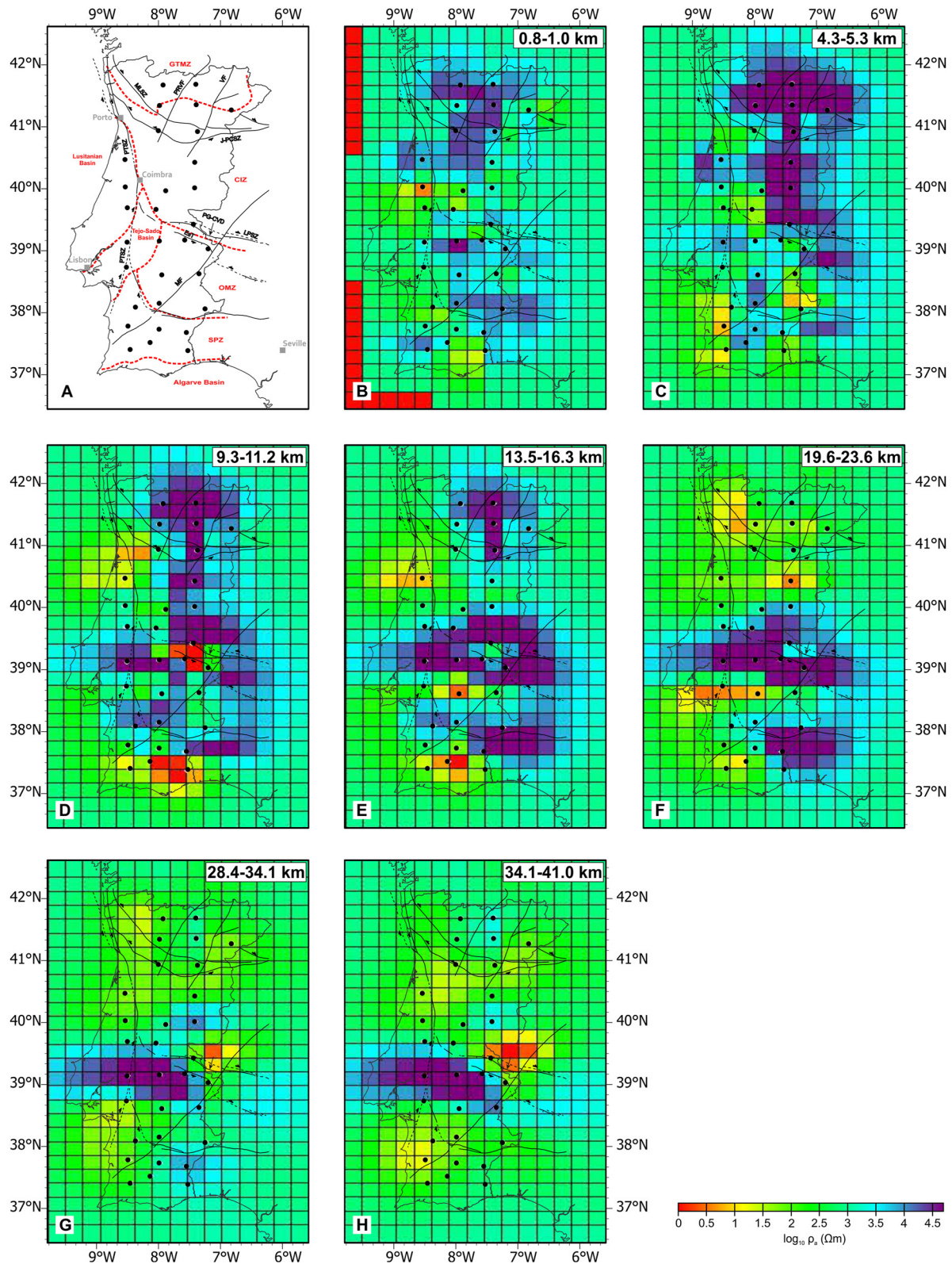


Figure 3. Horizontal slices of the B-S resistivity model for Portugal mainland. Black lines represent the major thrust faults in the area of study and black dots the location of the MT soundings. In panel A, dashed red lines separate the tectonostratigraphic zones based on Julivert et al. (1974). Abbreviations: CIZ—Central Iberian Zone; GTMZ—Galicia—Trás-os-Montes Zone; OMZ—Ossa-Morena Zone; SPZ—South Portuguese Zone; PTMZ—Porto-Tomar Shear Zone; VF—Vilariça Fault; PRVF—Penacova-Régua-Verin Fault; J-PCSZ—Juzbado-Penalva-Castelo de Vide Shear Zone; MLSZ—Malpica-Lamego Shear Zone; EsT—Espinhel Thrust; MF—Messejana Fault.

acquired between 1997 and 2002 with a 03E and GMS-06 systems (Almeida et al., 2005; Alves Ribeiro, 2018; Da Silva et al., 2007). The most recent MT soundings were taken during 2020-21, in the framework of the MAG-GIC project, using GMS-06, and GMS-07 systems. The electric field components E_x and E_y were measured in the N-S and E-W directions with a dipole length of approximately 100 m. The two horizontal magnetic field components B_x and B_y were acquired along the same directions as the electric field (Baltazar-Soares et al., 2021).

The magnetotelluric time series were processed using the manufacturer software MAPROS, based on the robust processing technique proposed by Egbert and Booker (1986), and only data with coherency over 0.6 (Friedrichs, 2007) were taken into account. MT data were inverted for the B-S conductivity model using the parallel version of the ModEM 3D code (Egbert & Kelbert, 2012; Kelbert et al., 2014; Meqbel, 2009). To include all the MT soundings, it was necessary to create a central uniform mesh of $25 \times 13 \times 58$ cells with 30 km width in the X , Y and downward directions, respectively. This central region was padded with 10 and 11 planes where cell sizes increase laterally by a factor of 1.3, in the X and Y directions, respectively. The vertical thickness is 25 m for the first layer; subsequent layer thicknesses successively increase by a factor of 1.2. In this way, the whole design mesh grid consisted of $45 \times 35 \times 58$ cells (see Figure S1 in Supporting Information S1). The ocean affects the MT impedance tensor (Parkinson & Jones, 1979; Mackie et al., 1988; F. A. M. Santos et al., 2001), thus a coarse 3D-bathymetry of the Atlantic Ocean and the Mediterranean Sea (Amante & Eakins, 2009) was included as a priori information, with a fixed resistivity of $0.3 \Omega \cdot \text{m}$ for seawater. A smoothing of 0.2 was applied along the X , Y , and Z directions.

Several inversions with different resistivity starting models were carried out. Due to the poor quality of some of the impedance diagonal coefficients, it was decided to consider only the off-diagonal coefficients of the impedance tensor with an error floor of 5%. The best fit between measured and predicted data, corresponding to a lower normalized root mean square error (nRMS) of 1.96, was achieved for a starting 3-layer model with a surface layer of $1,000 \Omega \cdot \text{m}$ until 100 km depth, followed by layers with $200 \Omega \cdot \text{m}$ until 2000 km depth and $1 \Omega \cdot \text{m}$ henceforth.

In Figure 3, horizontal slices of the B-S resistivity model for Portugal mainland are shown. It can be observed that the central and north regions of the territory are typified by a high resistive domain propagating in-depth, with resistivity values ranging from 3×10^3 to $3 \times 10^5 \Omega \cdot \text{m}$. To the west and south, roughly corresponding to the Portuguese western shore (Lusitanian and Tejo-Sado Basins) and the South Portuguese Zone (SPZ), respectively, the crustal resistivity tends to decrease unevenly to values ranging from 30 to $300 \Omega \cdot \text{m}$. The main resistivity anomalies identified in the MT inversion and shown in Figure 3 are discussed in Baltazar-Soares et al. (2021), in the light of the geological data and tectonic models that exist for Portugal mainland.

Forward sensitivity tests were performed in Baltazar-Soares et al. (2021) in order to estimate to which depth the inverted model is constrained by the MT data. This was done by replacing the final resistivity model computed through inversion, below a certain depth, with the resistivity value of the starting model at this depth (Cheng et al., 2021) and then computing the fit to the data. From these tests (see Figure S2 in Supporting Information S1), we can say that the B-S model is data-constrained only down to 40–50 km depth. The Moho layer in Portugal mainland varies from 28 to 32 km depth (Arroucau et al., 2021); thus, the B-S conductivity model allows us to retrieve information about the crust and the very upper mantle. However, to model GICs over one or more days (periods up to $\sim 10^4 - 10^5$ s), conductivity models need to represent the subsurface down to several hundreds of km (e.g., Boteler & Pirjola, 2017; Simpson & Bahr, 2005). In Section 6, we explain how we made use of local measurements of GICs at a particular substation to help constrain the nonuniqueness of the B-S conductivity model regarding the deeper layers of the mantle.

4. GIC Computation for the Whole Territory

The computation procedure follows a standard approach whereby GICs are considered quasi-direct currents, a reasonable assumption when comparing the GIC frequency range to the 50 Hz of phase currents flowing in high voltage power lines. In this way, only the resistive components of the power system impedances have to be considered. In particular, Lehtinen and Pirjola (1985) formulation of the circuit equations is applied, and GICs traveling along the groundings of transformers at substations are obtained by solving the matrix equation:

$$GIC = (1 + YZ)^{-1}J, \quad (2)$$

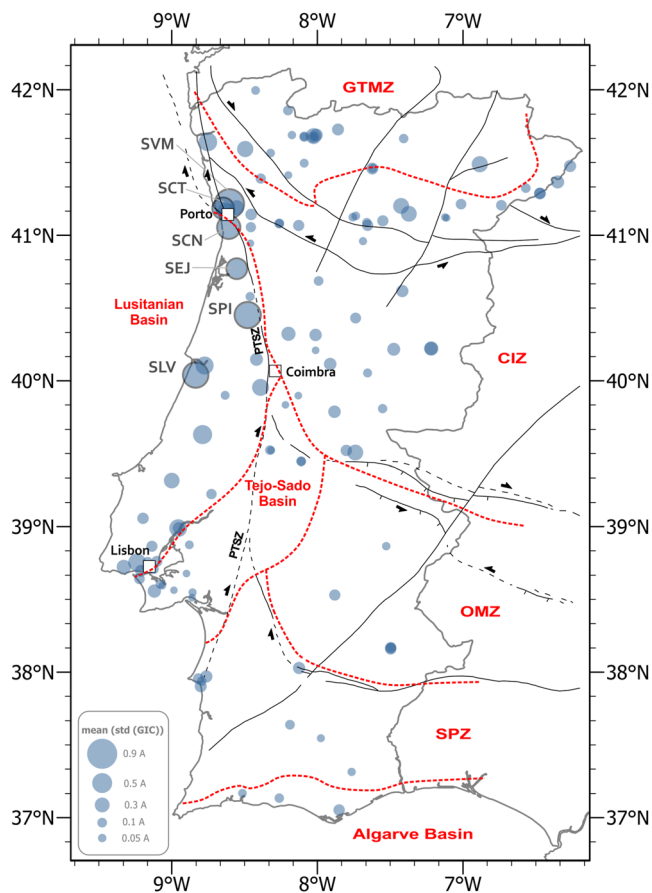


Figure 4. Mean of SD(GIC) values, computed from the 8 most intense geomagnetic storms of cycle 24. Substations with mean (SD(GIC)) > 0.5 A are highlighted with a dark contour. Black lines represent the major thrust faults in the study region. Dashed red lines separate different tectonic units. Abbreviations: GTMZ—Galicia-Trás-os-Montes zone; CIZ—Central Iberian Zone; OMZ—Ossa Morena Zone; SPZ—South Portuguese Zone; PTSZ—Porto-Tomar Shear Zone.

from estimations for SPI substation. This information is shown in Figure S3 in Supporting Information S1. As shown there and in Figure 4, SD(GIC) at SPI increases by a factor of about 3 during geomagnetic storms, up to ~1 A. For all storms, the highest values are found in the northern region, where the B-S conductivity model in Figure 3 displays higher resistivity values and/or higher resistivity contrasts, such as the Lusitanian basin/Central Iberian Zone (CIZ) transition. In the conductivity model, at ~10 km depth and below (Figures 3d–3h), it is seen an increase in electrical resistivity in the central region ($>10^4 \Omega \cdot \text{m}$), extending approximately W-E; however, contrasting to the north, this does not translate into an increase in the GIC values in this region. This can be explained by a concentration of line connections with longer line lengths in the northern region, mainly in the 400 kV, while there is a lack of lines in the central region (see Movies S1 and S2).

With respect to the effect of geomagnetic storms of cycle 24, it is observable that the substations with the highest GIC values are located in the vicinity of the Porto-Tomar Shear Zone (PTSZ). This shear zone separates the resistive CIZ from the conductive Lusitanian Basin and is therefore associated with a strong discontinuity in conductivity values on land. Besides, it is also close to another important conductivity contrast, the coastline. Thus, strong induced electric fields are expected to occur in this region, hence higher GIC values. The lowest mean SD(GIC) values are observed in the south region, where the conductivity is in general higher and without main contrasts on land, at least in the region where power lines lie (see Figure S4 in Supporting Information S1).

where I is the unit matrix, Y is the network admittance matrix, Z is the earthing impedance diagonal matrix with the substations' earthing resistances, and J is the column vector of perfect earthing currents (e.g., Boteler & Pirjola, 2017).

A total of 137 substations represent the power network grid of Portugal mainland, with 356 transformers and 334 line connections operating with three high-voltage levels of 150, 220, and 400 kV (see Figure 1). The longest transmission line in the circuit is 126 km in length, at 220 kV voltage level. 9 lines connecting to 6 substations belong to the Spanish power transmission system. These substations are integrated in the circuit in the same way as in Alves Ribeiro et al. (2021), that is, using the Equivalent Circuit model to represent neighboring networks (Boteler et al., 2013).

Using the B-S conductivity model derived for the studied region in ModEM, the surface impedance matrix was estimated in a regular grid. The electric induced field is computed in each grid point using the nearest observatory storm signal and then interpolated to the path points of the power lines using Delaunay triangulation. The effect of storm induced electromotive forces (emfs) in the calculation of GICs is considered through the elements of current J (see Equation 2). In our calculation, based on GEOMAGICA script (Bailey et al., 2017), the straight line approximation is used to model the power lines that connect two substations, and the path integral of the induced electric field to obtain the emf values is computed along these lines.

The impact of geomagnetic storms is often described using the maximum GIC value recorded/simulated at different transformer neutrals. However, this maximum value may not occur at the same time for different substations. Moreover, the maximum is not a statistically robust quantity as it is more susceptible to outliers. Instead, for each substation in the power grid (Figure 1), the standard deviation of simulated GICs during each storm in Table 1 were computed (see Equation 1). This requires the definition of a time interval for GICs duration, an issue discussed in Section 2.

The average and standard deviation of SD(GIC) values using the 8 storms were obtained. Figure 4 shows the distribution of the mean of SD(GIC) through all substations in the power network. To provide a reference level, SD(GIC) was computed during quiet days and average values 0.2–0.3 A were obtained

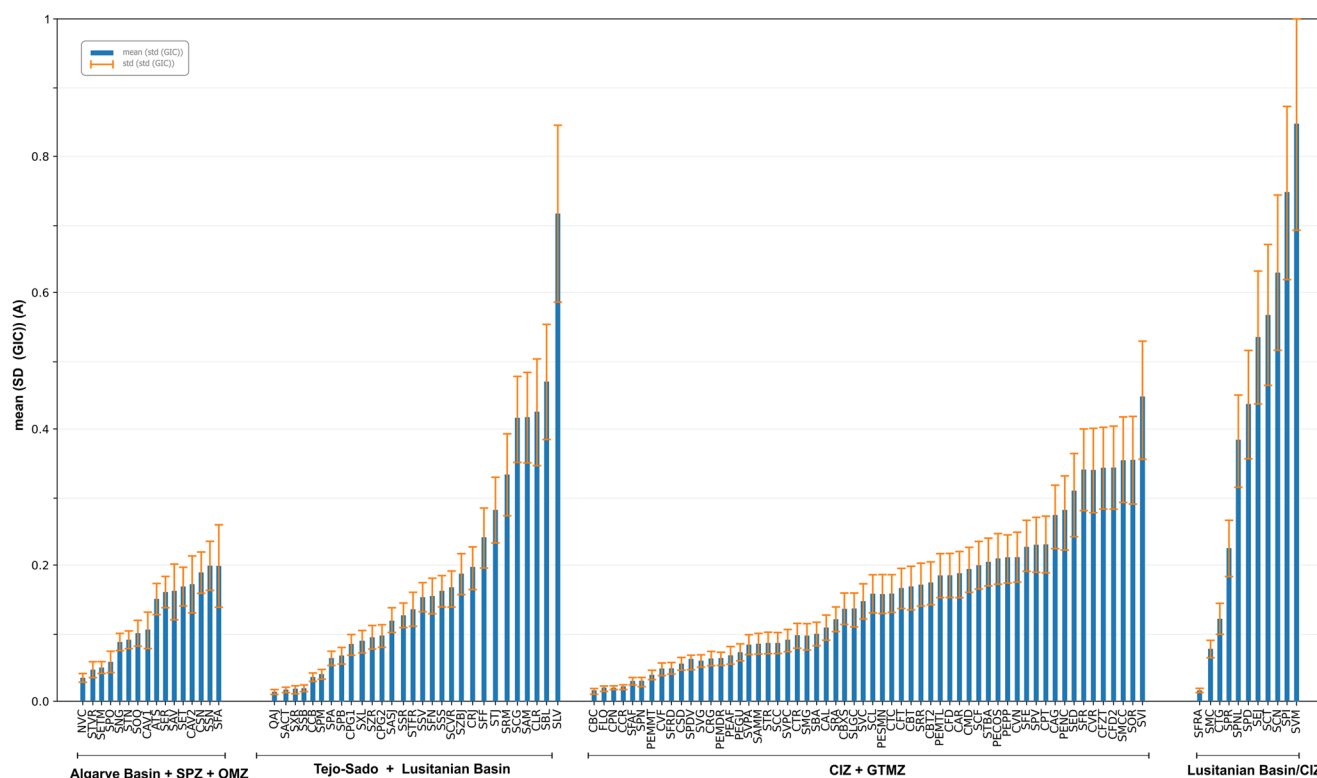


Figure 5. Distribution of the mean of SD(GIC) values (blue bars) over substations grouped into four large geotectonic units, for the 8 most intense geomagnetic storms of solar cycle 24. Orange bars for the standard deviation of SD(GIC) values. Abbreviations: GTMZ—Galicia-Trás-os-Montes zone; CIZ—Central Iberian Zone; OMZ—Ossa Morena Zone; SPZ—South Portuguese Zone.

Figure 5 shows the distribution of mean SD(GIC) values (and corresponding standard deviations) for different substations grouped into four characteristic geologic units: “Algarve Basin + SPZ + OMZ” and “CIZ + GTMZ,” represent two regions without main tectonic contrasts, the first one to the south (lower electrical resistivity and lower SD(GIC) values) and the latter to the north (higher electrical resistivity and higher SD(GIC) values). A third group gathers substations in the “Tejo Sado + Lusitanian Basins,” a region of relatively high conductivity on land down to ~10 km depth and relatively close to the coast. The “edge effect” at end-of-line substations (Pulkkinen et al., 2017; Zheng, Liu, et al., 2013) can be seen in Figure S5 in Supporting Information S1, especially for a uniform induced electric field along the W-E direction. Finally, the fourth group “Lusitanian Basin/CIZ” gathers substations which are simultaneously close to the coast and to the PTSZ. This last group clearly shows the highest SD(GIC) values. Although the relatively high number of 400 kV lines in this region contributes to these results, as well as the “edge effect” typical of end-of-line substations, these features alone do not explain the GIC distribution pattern (see Figure S5 in Supporting Information S1 for the isolated effect of the parameters and topology of the grid). Instead, the high GIC values in this region could be due to the cumulative effect of two conductivity discontinuities (the coastline and the PTSZ), where the geoelectric field is expected to suffer important distortion (e.g., Marshall et al., 2019).

Analyzing GIC results from the perspective of the grid topology, we come to the same conclusions that have been evidenced by other authors: higher GICs are associated with connections to the highest voltage (400 kV), to longer lines and a higher number of total line connections (e.g., Zheng, Boteler, et al., 2013). The added contribution of these three different factors can be seen at substation SLV (see Figure S4 in Supporting Information S1), connecting to 6 different lines, all at 400 kV, and among which lie the longest lines in the grid with 111 and 87 km length.

5. Validation of GIC Computations

GIC estimated values might not be accurate because they rely on parameters and assumptions with associated uncertainties and need to be validated by measurements. An acquisition device for monitoring GICs was

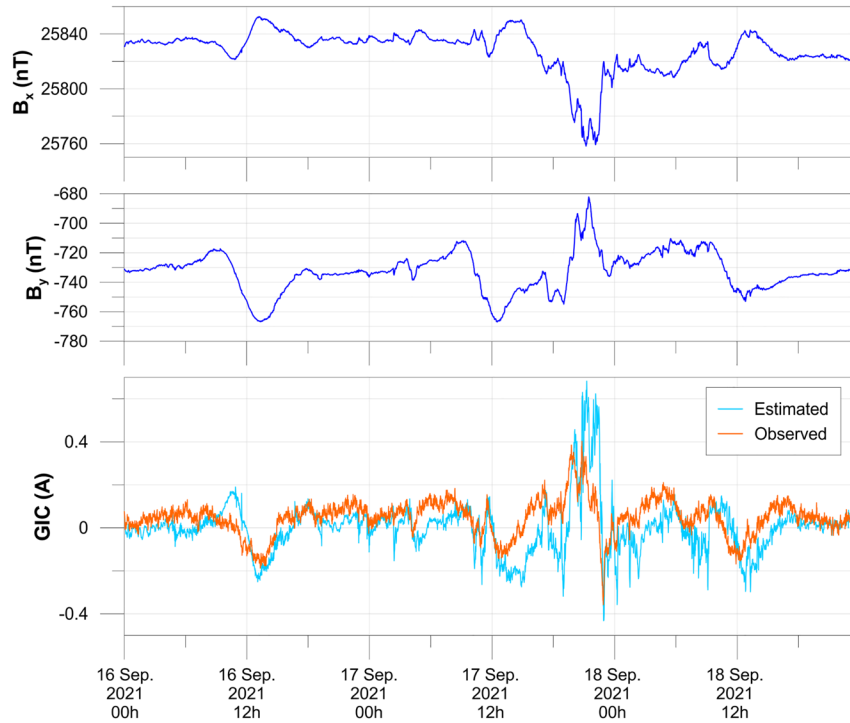


Figure 6. Geomagnetic storm of 17 September 2021. Geomagnetic horizontal components B_x and B_y at COI observatory. Estimated and observed geomagnetically induced currents at the TFR6 transformer neutral in Paraimo during the same period.

assembled, calibrated, and tested in the laboratory (R. R. Santos, 2021). The instrumental setup consists of a Hall effect current sensor LEM HOP 1000-SB with a manufacturer's sensitivity 4 mV/A, and a Raspberry Pi 4 Model B platform with a high resolution 24-bit digitizer board (Waveshare AD/DA). The system comprises a mobile internet router. Data is stored in the local database and can be accessed through a remote connection. Laboratory tests revealed a true sensitivity of 4.25 mV/A, a current offset of -0.39 A at constant (ambient) temperature and a noise standard deviation of 0.056 A.

On the 30th of August 2021, the acquisition device was successfully installed and set in operation at Paraimo substation (SPI), about 35 km north of Coimbra (Figure 1) in order to provide observational constraints for estimated GIC values in the national power network. The Hall effect sensor was clamped around the neutral grounding cable of the TRF6 transformer.

Figure 6 shows the comparison between the GIC recorded at Paraimo during the 17 September 2021 storm ($Dst(\min) = -64$ nT) and the GIC model estimate. The two curves present similar temporal variations with a Pearson correlation coefficient of 0.60. Differences are apparent, nonetheless, in Figure 6, namely a phase shift at certain occasions, as just after noon of 17th and 18th of September, when measurements follow more closely the B_y than the (minus) B_x curves of the geomagnetic field components at COI. This behavior suggests that the impedance tensor, which is a transfer function relating the geomagnetic to the induced electric fields, might need to be improved. This could be associated with imprecisions in the B-S conductivity model, as will be discussed ahead.

By separating the observed/estimated signals into three distinct frequency bands, it is possible to more accurately characterize the observed behavior. Butterworth filters (Butterworth, 1930) are used in signal processing as they provide a flat frequency response for the desired passbands while avoiding the drawbacks of window filters as the presence of ripples. Butterworth filters include low-pass filters, for which the filter response is defined as:

$$R_{low} = 1 / (1 + C(\omega/\omega_0)^{2n}), \quad (3)$$

and high pass filters with a response defined as:

$$R_{high} = 1 / (1 + C(\omega_0/\omega)^{2n}). \quad (4)$$

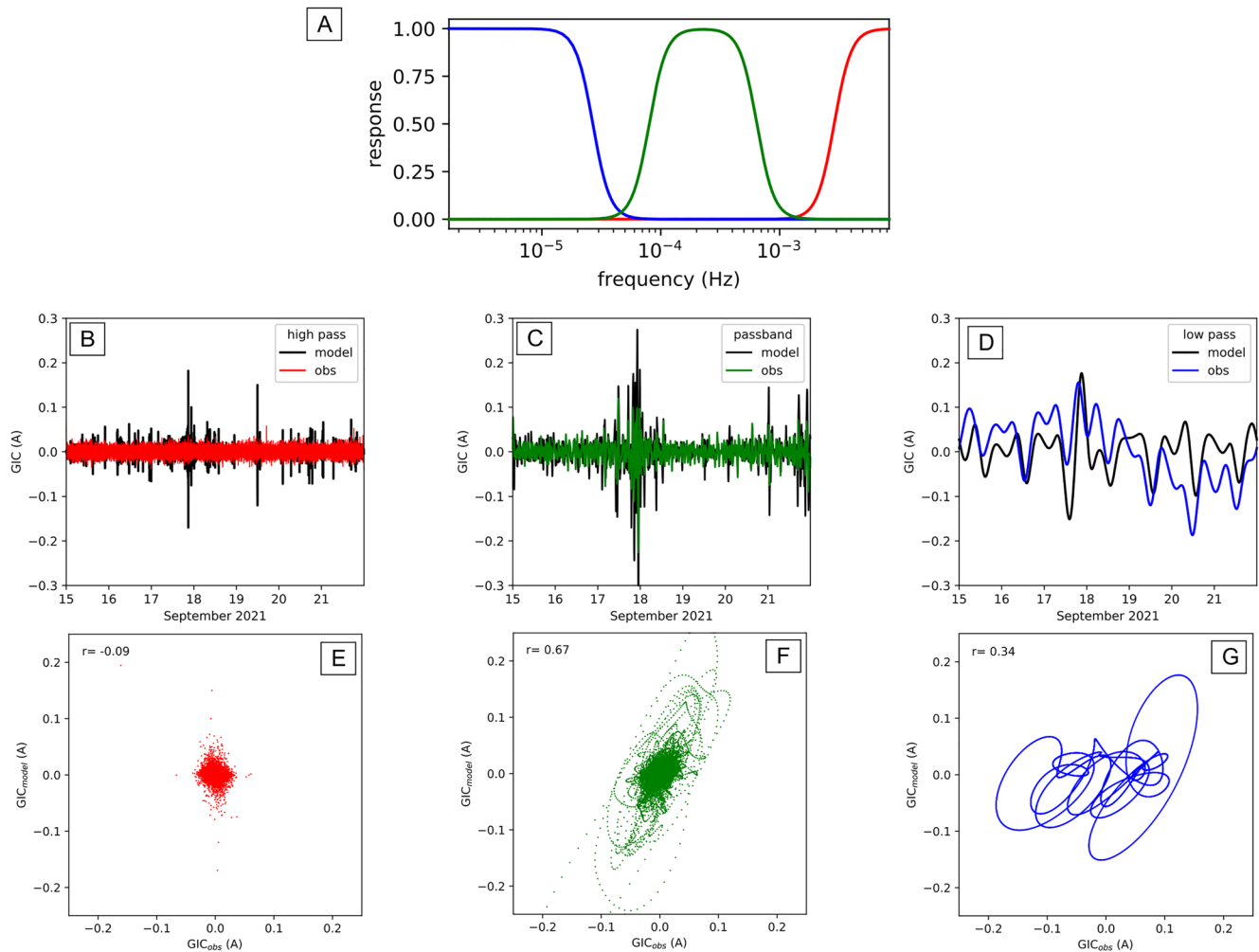


Figure 7. The geomagnetic storm of 17 September 2021, at Paraimo. Comparison between observed and simulated geomagnetically induced current signals in different frequency bands, after using a high-pass (b), (e), band-pass (c), (f) and low-pass (d), (g) Butterworth filters (a).

In both cases, n corresponds to the filter's order, which in the present work is set to 3. The constant C was chosen so that at the cut-off frequency ω_0 the filter response is $R(\omega_0) = 1/\sqrt{2}$. This gives $C = \sqrt{2} - 1$. Band-pass filters are nothing more than a combination of low-pass and high-pass filters. For this analysis, the cut-off frequencies considered are $1/300$ Hz for the high-pass filter, $1/3$ hr $^{-1}$ and 2 hr $^{-1}$ for the band-pass filter, and 2 days $^{-1}$ for the low-pass filter (Figure 7a). In the high frequency band (Figure 7a, in red) the sensor signal appears to be uniform noise with amplitude ~ 0.02 – 0.03 A (Figures 7b and 7e). This is related to the sensor detection limit in this frequency band. The correlation between model and observations in the low frequency band is low (Figures 7a, 7d, and 7g). This is probably due to inaccuracies in the conductivity model concerning deep layers. In fact, MT soundings used in the inversion for the B-S conductivity model can not resolve these layers properly due to constraints imposed to the sampled frequency range. The uncertainty in the conductivity model is the main limitation of our calculations, as discussed in the following section, for which long period MT soundings could be a solution. Finally, in the intermediate pass band (Figures 7a, 7c, and 7f), the relationship between model and observation is quite good, with a correlation of 0.67, a result that we consider very encouraging.

6. Possible Sources of Uncertainties

6.1. Conductivity Model

The B-S conductivity model used in this study has some limitations. MT soundings were acquired with broadband stations (10^{-3} s to 10^3 s), while the period range of interest in geomagnetic storms for the purpose of GIC

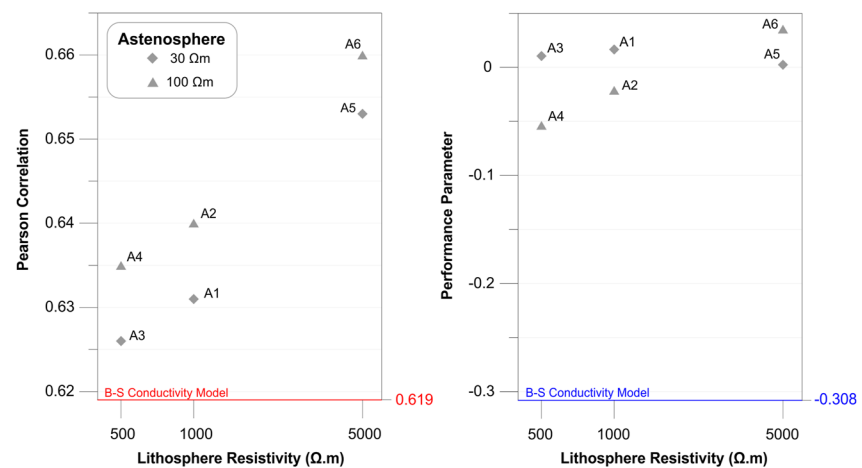


Figure 8. Pearson correlation coefficient (R) and performance parameter (P), comparing the estimated and observed geomagnetically induced currents at SPI on the 17th September 2021. Different values refer to different conductivity models, changing the lithosphere (from 40 to 100 km depth) and asthenosphere (from 100 to 2,000 km depth) resistivities.

studies extends to 10^4 – 10^5 s (Boteler & Pirjola, 2017; Simpson & Bahr, 2005). As a result, information on the electrical conductivity from deep layers is missing or is incomplete. In fact, the analysis of the GIC sensor data at SPI revealed a smaller correlation coefficient between observations and estimations for the lower frequency range, with an irregular phase lag, thus reinforcing the inadequacy of the model at larger depths (Figures 7d and 7g).

Several tests were performed to evaluate the influence of the electrical properties of the lithosphere and asthenosphere, by changing the resistivity of these two layers in the 3D conductivity model while keeping the same parameters as in the B-S model down to 40 km depth. The direct problem was solved to compute the resulting GICs, and the simulations were compared with observations at the SPI substation. In Figure 8 is represented the Pearson correlation coefficient, R , between the observed and estimated GIC at SPI, for the geomagnetic storm on the 17 September 2021. The other computed statistics is the performance parameter, P (Torta et al., 2014). Like R , P is defined to be less than 1 and to get closer to 1 as model predictions approach observations. However, P allows testing the similarity of the curves' amplitudes, while R tests only the similarity of their shapes. Values were computed for observed and simulated series removed from their high frequency content (see Equation 4), as the sensor cannot resolve this part of the signal. For the two asthenosphere resistivity values tested (30 $\Omega\cdot\text{m}$ and 100 $\Omega\cdot\text{m}$), the correlation coefficient tends to increase for higher values of the lithosphere resistivity (Figure 8). The best correlation coefficient is obtained for a more resistive asthenosphere underlying a very resistive lithosphere (model A6). In this case, not only the Pearson correlation coefficient is (slightly) higher, but the performance value P also attains the best score (Figure 8). The P parameter appears to be more sensitive than R to changes in the GIC model, when testing different conductivity profiles. As an example, comparing results obtained with A5 and A6 models, we can see how a variation of a factor of about three in the conductivity of the asthenosphere leads to changes of about 0.04 in P , but less than 0.01 in R .

A possible first-approach explanation for the high resistivity values in depth that SPI data seem to require might be found in the tectonic models for this region. SPI substation is located in the vicinity of PTSZ, separating the Lusitanian Basin from the CIZ. This region reveals evidence of successive geotectonic events since the Variscan-Late Variscan and still suffered reactivations during the Alpine cycle (Chaminé et al., 2003; Pereira et al., 2010). It is important to carefully consider the requirement for a very high resistive lithosphere in model A6. We may not completely disregard the possible errors in the conductivity model for the first 40 km due to a complex lateral heterogeneity (where granitoids may play a role) that our model can not reproduce. Due to the nonuniqueness of the problem, and since we fixed the model for the first 40 km, those errors could be partially solved by increasing the resistivity at deeper layers.

The blind variation of layer resistivities seems to point toward the need for larger resistivity values in depth and, consequently, the participation of deeper layers of the solid Earth in the induction process in order to better explain the observations at Paraimo. Nonetheless, the observed changes shown in Figure 8 do not represent a

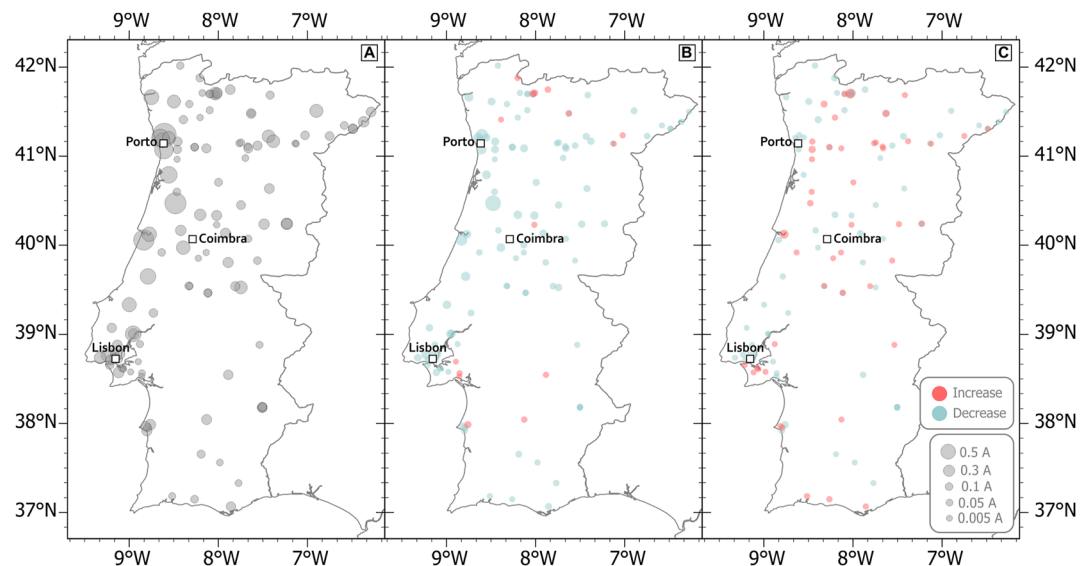


Figure 9. For the geomagnetic storm of 17 March 2015, (a) the distribution of SD(GIC) in the power network, and differences in SD(GIC) when replacing (b) the B-S conductivity model for A6; (c) the standard grounding resistance of non-REN substations for values of the nearest REN substations inside the same geological unit.

significant improvement. In order for a more judicious correction of the conductivity model at greater depth, long-period MT soundings must be acquired in the future. It should also be kept in mind that a 50×50 km grid of MT soundings might cover Portugal mainland large scale features reasonably; however, due to the complex geology of the territory, it might not capture important local structures. Murphy et al. (2021) showed that a denser MT grid can lead to differences up to an order of magnitude when estimating the geoelectric hazard.

Other studies that used GIC measurements to improve the imaging of the subsurface conductivity, include the MT-type inversion by Pulkkinen et al. (2007), using a 1D assumption. Such assumption is particularly suited for the low frequency GIC signal that penetrates deeper in depth, and the method can be tested in the future using GIC measurements at SPI and geomagnetic observations from the nearby COI observatory.

We estimated uncertainties in GIC predictions at all substations due to the nonuniqueness of the conductivity model for the St. Patrick storm (17/03/2015). These estimations were computed from differences between SD(GIC) values obtained with the B-S conductivity model and the A6 conductivity model (Figure 9b). Using model A6, it is seen a general decrease in the SD(GIC) values by a scale factor of about 0.6, on average. This systematic decrease of SD(GIC) is consistent with a decrease of the resistivity value for the asthenosphere, from $200 \Omega \cdot \text{m}$ (B-S model) to $100 \Omega \cdot \text{m}$ as in A6. The strongest decrease in SD(GIC) is observed at SPI, with a value about 0.5 A (Figure 10a), corresponding to a decrease of the GIC peak of 3.6 A.

6.2. Uncertainties in Grounding Resistances

There still remain some unknown parameters of the power network concerning the substations not managed by REN. Transformers of non-REN substations are treated as Y-Delta transformers, with Y connections from the side of the REN network. They are integrated in the circuit using equivalent values for Y-Delta transformers wiring resistances. However, the information on grounding resistance values for these substations is not available. In the GIC model used in this study, it was assumed 0.2Ω for the grounding resistance of all non-REN substations (see Figure S6 in Supporting Information S1).

As it can be seen from Figure 3b, the conductivity distribution at the near-surface differs significantly from south to north, being more resistive in the north. The transmission power lines operators need to adapt the grounding resistance values to the local electrical properties. This is expected to happen also for non-REN substations, with higher values in the north than in the south. As a test to expose the effect of using non-realistic grounding resistances, it was used for each non-REN substation the grounding resistance of a REN substation located in the vicinity, inside the same geological unit. The north region is where the test grounding resistances differ the

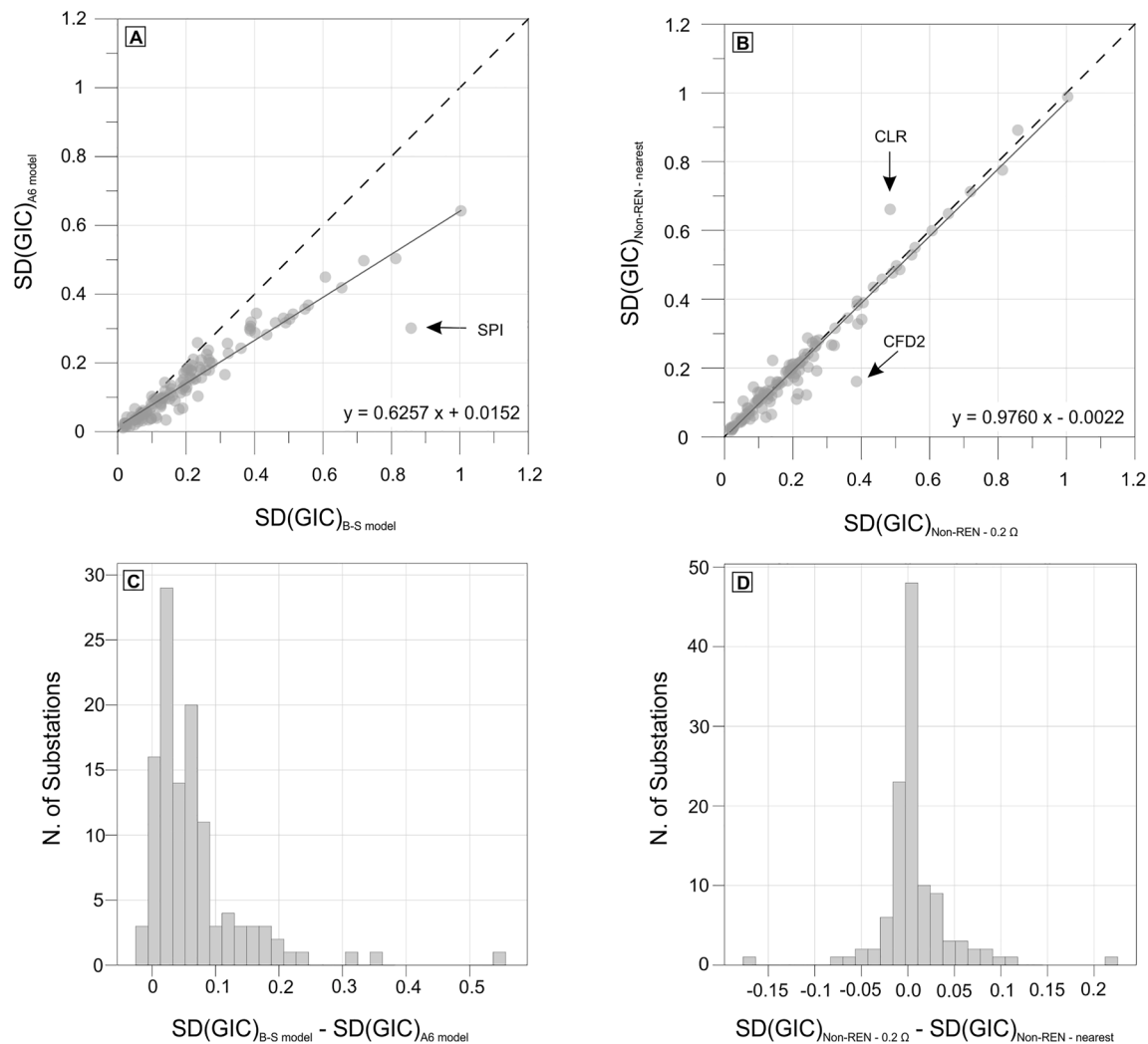


Figure 10. For the geomagnetic storm of 17 March 2015, the distribution of SD(GIC) differences in scatter plots (top) and histograms (bottom), when replacing (a) and (c) the B-S conductivity model for A6; (b) and (d) the standard grounding resistance of non-REN substations for values of the nearest REN substations inside the same geological unit.

most from the standard value of 0.2Ω assumed in the model, reaching values up to 0.8Ω higher (see Figure S6 in Supporting Information S1).

For the St. Patrick storm (17/03/2015), we estimated uncertainties in GIC predictions at all substations due to the missing information regarding the non-REN substations earthing resistances. When comparing the SD(GIC) differences between our GIC model (B-S conductivity model and 0.2Ω values for non-REN earthing resistances) with the replacement of the non-REN earthing resistances with values more in agreement with the local geology (Figure 9c), the peak GIC differences are especially high, as expected, at non-REN substations, reaching around 2 A in the northern part of the network. The CLR substation stands out, since we observe an increase of 36% of the SD(GIC) (Figure 10b) for which contributes mainly the decrease to 0.1Ω of the local earthing resistance. In the case of CFD2, the earthing resistance increases to 1Ω , leading to a 60% reduction of the SD(GIC). As expected, the grounding also indirectly affects other substations, where the largest differences in peak GIC values correspond to a decrease of 0.6 A (SRA) and an increase of 0.3 A (STN). On average, however, the SD(GIC) differences due to uncertainties in earthing resistances are smaller than those due to conductivity model uncertainty, and there is no significant systematic variation.

Finally, Figures 10c and 10d show the statistical distribution of SD(GIC) differences when testing two different conductivity models and two different sets of earthing resistances. In the case of GIC computation for the

Portuguese transmission power network, where most of the direct circuit parameters were provided by REN, the lack of a more precise subsurface conductivity model is causing the most important limitations to precise estimations. As we can see, different possible conductivity models lead to significant changes in the SD(GIC) values at different substations, with a scaling factor of ~ 0.6 (i.e., of the order 40%) for the case of the two models tested in this study. Differences in SD(GIC) due to grounding resistances are only of the order of 2%.

7. Discussion and Conclusions

In this article, we present the first GIC computations for the entire Portuguese mainland high voltage power network (400, 220, and 150 kV) based on realistic local properties. Two main improvements are brought to the model, in comparison to previous work (Alves Ribeiro et al., 2021), allowing to estimate the geomagnetically induced electric fields and the related driven GICs with higher accuracy:

1. A new 3D conductivity model (the B-S model), inverted from 31 MT broadband soundings covering the entire national territory;
2. Information on the circuit parameters for the full power transmission grid.

Besides, we report the first GIC measurements on the grid at the Paraimo (SPI) substation, which provides a local test of the model. Comparison of GIC model estimates with data at this substation show encouraging results.

GIC computations rely on local geomagnetic observations for the eight strongest geomagnetic storms during the solar cycle 24. Based on obtained GIC estimations, it was possible to identify the critical substations in the Portuguese power network which are prone to strongest GIC effects. In fact, the maximum SD(GIC) of 1.2 A (for the geomagnetic storm of October 2013) and the peak GIC value of 15.1 A (for the June 2015 storm) were both obtained at Vermoim substation (SVM), located in the northern limits of the shear zone PTSZ, and also next to the coast. In general, the highest GIC estimations are found in the northwest of the territory. The combination of two factors may justify these results: the proximity both to a strong conductor, the Atlantic Ocean, and the PTSZ. The latter is a region that presents strong contrasts in resistivity, as it separates the low-resistivity sedimentary Lusitanian Basin from the high-resistivity granitic region of Central Iberia (CIZ).

Girgis et al. (2020) defined the degree of total susceptibility to the effects of GICs, taking into account both the transformer design and the magnitude of GICs. For GIC intensities below 15 A/phase, the GIC exposure level was considered low. We can then conclude that, for geomagnetic storms of intensity similar to the strongest events of solar cycle 24 (a particularly weak cycle), the transformers in the Portuguese power network have a low exposure level and are not at risk due to GICs.

In the last 4 months of 2021, we had access to observations from a Hall effect sensor installed on a power transformer at the Paraimo substation in the vicinity of Coimbra. The minor events recorded still allowed us to test the new GIC model. Comparing GIC observations with the simulations for transformer TRF6, the correlation coefficient between the two time series allows us to discard the null hypothesis and is relatively high. In search for a better characterization of remaining differences, filters (in particular Butterworth filters) were used to separate data (and simulations) into different frequency bands. For the relatively weak events recorded in the study period, observations are dominated by instrumental noise (amplitude ≈ 0.02 – 0.03 A) in the high frequency regime. Only for stronger geomagnetic storms, or by using a sensor with a higher sensitivity, will it be possible to evaluate the model performance in this frequency domain. Differences between observations and simulations on the low frequency end of the spectrum, in particular an apparent phase shift of a few hours, suggest that we do not have adequate knowledge of the resistivity values at larger depths. This is first due to the fact that the used MT soundings do not reach that deep.

Two sources for uncertainties in GIC estimations were considered. Testing variations to the conductivity model at the lower layers (lithosphere and asthenosphere) suggests that a mere 1-D model is too simple and does not account for deviations between observations and simulations at Paraimo; this might be due to the geological complexity in this particular region (located in the vicinity of two very different geological units separated by the PTSZ). Even if not conclusive (different conductivity values at depth display similar degrees of correlation and global performance with respect to the observations), the results of our tests point toward higher resistivity at the lower layers. The other test that was done reinforces the importance of having the true values of the grounding resistances. When considering the grounding at a non-REN substation (which is not provided) to be the value of

the closest REN substation in the same geological structure, we could estimate the error that is being done. In the particular case of the Paraimo substation, the differences assuming one or the other scenarios are smaller than the instrumental errors of the Hall effect sensor installed there. Yet that is not the case for some substations. Results from our tests show that uncertainties due to the unknown deep conductivity amount on average to 40% of estimated SD(GIC), while uncertainties due to the unknown earthing resistances are of the order of 2% on average.

A realistic conductivity model for the underlying ground is the main asset in the accurate calculation and prediction of GICs. Like other groups, we have been making efforts toward improving our knowledge of local electrical properties with new MT campaigns. Long-period MT surveys could bring new insight into the present lack of information on the conductivity model at higher depths. Also, at the surface, the more precise the conductivity model, the more accurate the computation of the electric field path integrals will be and, as a result, GIC estimation becomes more accurate. Therefore, we plan to acquire more MT soundings within a tighter grid.

Meanwhile, GIC monitorization at additional substations can be used to constrain the conductivity models, as we have shown here. GIC monitorization is also central to test the network model and possible errors in the electric circuit parameters, as for example, grounding resistances. For this reason, we are currently working on developing a GIC measuring system with higher sensitivity and plan to install sensors in a larger number of substations, in different geological contexts, for further control.

Data Availability Statement

INTERMAGNET data for SPT and SFS can be found at <https://intermagnet.org/data-donnee/download-eng.php> and WDCG data for COI are found at <https://wdc.bgs.ac.uk/catalog/master.html>. MT sounding data used to invert for the conductivity model in Figure 3 are found at <https://doi.org/10.5281/zenodo.7147543>, the GIC statistics used to produce Figures 4 and 5 are found at <https://doi.org/10.5281/zenodo.7443212> and the GIC measurements and estimates at the Paraimo (SPI) substation used to produce Figures 6 and 7 are found at <https://doi.org/10.5281/zenodo.7446370>.

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